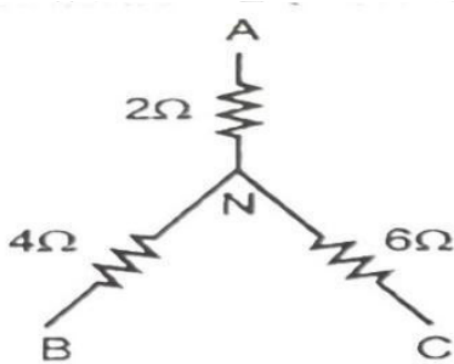


Q1

(a) Convert the star circuit into its equivalent delta circuit.



(b) Write down the relation between:

- Line current and phase current
- Line voltage and phase voltage
- Power in a balanced three-phase delta-connected balanced load

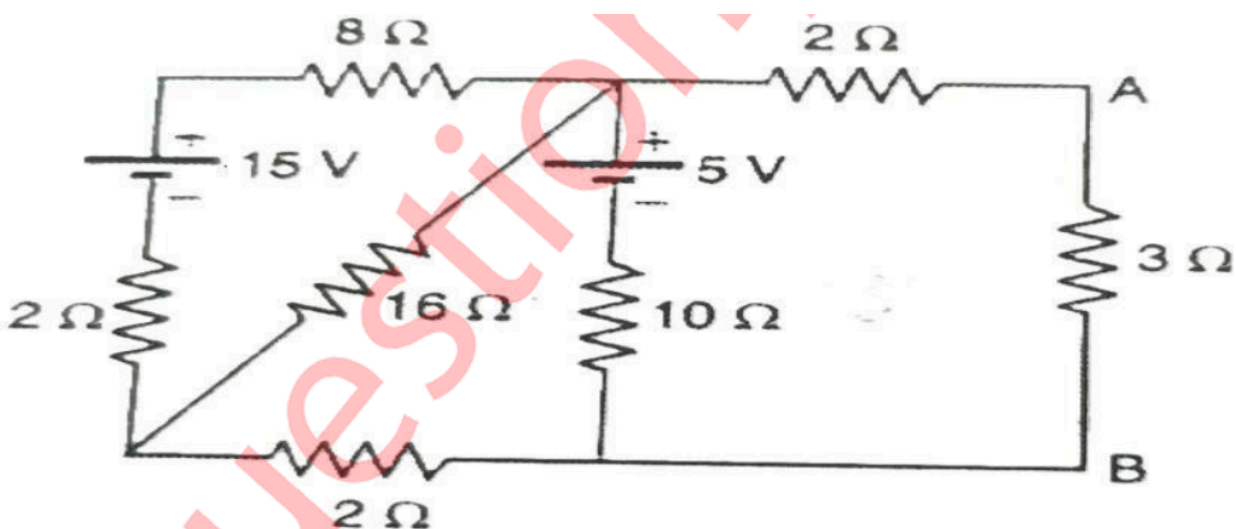
(c) Explain the concept of back EMF in a DC motor.

(d) What are the losses in a single-phase transformer?

(e) Draw and explain the construction diagram of an LED and enlist its applications.

Q2

(a) Using Norton's theorem, calculate the current through the 3Ω resistor across terminals A and B in the given circuit.



(b) Explain the working principle of a three-phase induction motor and mention its types.

